

- 3 A ball of mass m kg is projected vertically upwards with initial speed U m s⁻¹ and moves under gravity. At time t s after projection, the ball has travelled a distance x m and its speed is v m s⁻¹. There is a resistive force of magnitude mkv^2 N, where k is a positive constant.

(a) Show that the distance travelled by the ball when it is moving upwards is $x = \frac{1}{2k} \ln \left(\frac{g + kU^2}{g + kv^2} \right)$. [4]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

(b) Find the time taken for the ball to reach its maximum height.

[4]

[illegible]